

Programmable Logic Controller Gateway Implementation Guidelines

PL12-500

Implementation
PLC Gateway

***Programmable Logic
Controller Gateway
Implementation Guidelines***

PL12-500
Release 500
9/97

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Revision 02 – September 30, 1997

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About This Publication

This publication summarizes the Programmable Logic Controller Gateway (PLCG) implementation process, guides you to procedures and references you need to implement PLCGs, defines the hiway and box/slot entities you must build for each PLCG, and describes PLCG operation considerations in implementing and using PLCGs.

This publication supports **TotalPlant** Solution (TPS) system network Release 500. TPS is the evolution of TDC 3000^X.

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INTRODUCTION

Section 1

This section summarizes the Programmable Logic Controller Gateway implementation tasks, lists publications that you will refer to in order to implement PLCGs, and describes implementation dependencies.

1.1 SUMMARY OF PLCG IMPLEMENTATION TASKS

While most of the the information in this publication relates to Programmable Logic Controller Gateway (PLCG) functions, PLCG data points, and PLCG operating considerations, other implementation activities must also be completed to make the PLCG functional. The Engineering Personality activities listed below may be affected by the implementation of a PLCG or must be used to implement a PLCG.

See subsection 1.2 for references to instructions for each of the activities listed below.

Activities named in **THIS TYPEFACE** are activated by targets on the Engineering Personality Main Menu.

- **UNIT NAMES**—The process units defined for each PLCG data point are established in this activity.
- **AREA NAMES**—The area name and descriptor for any units with PLCG points that are assigned to an area are established in this activity.
- **LCN NODES**—Identifies and defines the nodes on the Local Control Network. In the case of PLCGs, this activity defines the node numbers for the PLCGs and the process network number for the PLCGs' emulated Data Hiways. Note that because the PLCG is supported by the Hiway Gateway software, you configure PLCGs as if they were HGs.
- **VOLUME CONFIGURATION**—The PLCG (HG) checkpoint volume(s), &7np is established in this activity. Volume &7np must have adequate storage space to accommodate the PLCG checkpoint data, including the data for its emulated DHPs (boxes).
- **APPLICATION MODULE**—Any AM points that are members of a control strategy that includes PLCG points are built in this activity. Connections to the PLCG points are defined in `tagname.parameter` form.
- **HIWAY GATEWAY**—The Hiway point that defines the PLCG's emulated Data Hiway and the Box/Slot points that define the functions and content of the emulated DHPs are built in this activity. Also, the PLCG data points are built in this activity.
- **PICTURE EDITOR, FREE FORMAT LOGS, BUTTON CONFIGURATION**—Any of the pictures, logs, and buttons built by these activities can access PLCG points, once the points are built and loaded.

- **HM HISTORY GROUPS**—PLCG data point values for which continuous history is to be collected are defined in this activity by assigning them to specific HM history groups.
- **AREA DATA BASE**—This activity defines how and where data for data points, including PLCG data points, are used and displayed in a given process area. The area database is the database loaded into a Universal Station, so that the database defines the process area monitored and controlled through the US.
- **Control Language (CL)**—The PLCG does not have a CL program feature, but CL/AM programs can read from and write to PLCG data points.
- **Ladder Logic Programming**—The make and model of the PLC(s) connected to the PLCG can widely differ, so the Honeywell IAC publications do not provide information for ladder logic programming. Consult the manuals provided by the PLC vendor(s).

1.2 REFERENCES

1.2.1 References for Engineering Activities

- **UNIT NAMES and AREA NAMES**

Network Form Instructions in the *Implementation/Startup & Reconfiguration - 1* binder.

Network Data Entry in the *Implementation/Startup & Reconfiguration - 1* binder.

- **VOLUME CONFIGURATION**—Section 7 of the *Engineer's Reference Manual* in the *Implementation/Startup & Reconfiguration - 2* binder. For the PLCG checkpoint volume(s), follow the instructions for HG checkpoint volumes.

- **APPLICATION MODULE**

Application Module Control Functions in the *Implementation/Application Module - 1* binder.

Application Module Algorithm Engineering Data in the *Implementation/Application Module - 1* binder.

Application Module Parameter Reference Dictionary in the *Implementation/Application Module - 1* binder.

Data Entity Builder Manual in the *Implementation/Engineering Operations - 1* binder.

- **HIWAY GATEWAY (PLCG)**

System Control Functions in the *Implementation/Startup & Reconfiguration - 2* binder.

Programmable Logic Controller Gateway Control Functions in the *Implementation/PLC Gateway* binder.

Programmable Logic Controller Gateway Parameter Reference Dictionary in the *Implementation/PLC Gateway* binder.

Data Entity Builder Manual in the *Implementation/Engineering Operations - 1* binder.

- **PICTURE EDITOR,**

Instructions for reference, data entry, and form instructions for the Picture Editor are in the *Implementation/Engineering Operations - 2* binder.

- **HM HISTORY GROUPS, FREE FORMAT LOGS, BUTTON CONFIGURATION**

Instructions for these activities are in the *Implementation/Engineering Operations - 1* binder.

Data Entity Builder Manual in the *Implementation/Engineering Operations - 1* binder.

- **AREA DATA BASE**

Area Form Instructions in the *Implementation/Engineering Operations - 1* binder.

Data Entity Builder Manual in the *Implementation/Engineering Operations - 1* binder.

1.2.2 References for Hardware Implementation

1.2.2.1 Site Planning

LCN Site Planning manual in the *System Site Planning* binder.

Programmable Logic Controller Gateway Planning, Installation, and Service in the *Implementation/PLC Gateway* binder.

1.2.2.2 Installation and Checkout

Programmable Logic Controller Gateway Planning, Installation, and Service in the *Implementation/PLC Gateway* binder.

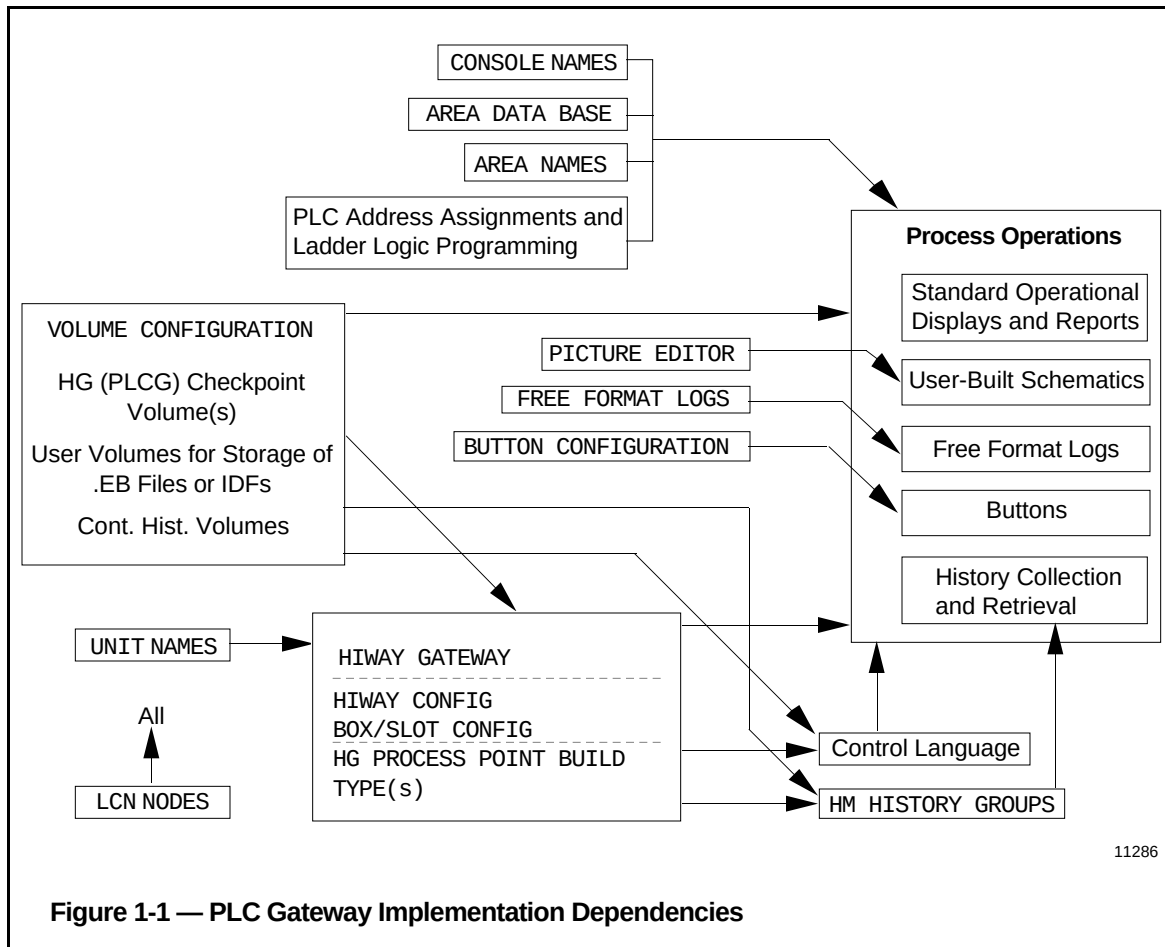
1.2.2.3 Service

Five/Ten-Slot Module Service in the *LCN Service - 2* binder (Programmable Logic Controller Gateway).

Programmable Logic Controller Gateway Planning, Installation, and Service in the *Implementation/PLC Gateway* binder.

1.3 PLCG IMPLEMENTATION DEPENDENCIES

Figure 1-1 shows which PLCG implementation tasks depend on information entered in other tasks. This figure does not necessarily dictate the order in which the tasks must be completed, but it does show all dependencies that must be satisfied before the PLCG can be fully operational.



HIWAY AND BOX/SLOT CONFIGURATION Section 2

This section provides guidelines for building hiway and box/slot configuration entities for each PLC Gateway.

2.1 PURPOSE OF THE HIWAY AND BOX/SLOT ENTITIES

Each hiway entity (point) defines for the LCN system, the emulated Data Hiway. Each box/slot entity defines for the LCN system, the HG (PLCG) box(es), or one of the emulated Data Hiway Ports (DHPs) in the PLCG and each slot in each DHP.

2.2 HIWAY AND BOX/SLOT ENTITIES TO BE BUILT

The PLCGs in a PLCG pair always use hiway addresses 02 and 03 on the emulated Data Hiway. The emulated DHPs in each PLCG or PLCG pair always use hiway addresses 08 through 15. For more information about this addressing scheme, refer to subsection 2.5 in *PLC Gateway Control Functions* in the *Implementation/PLC Gateway* binder.

2.2.1 Hiway Entities

You must build a hiway entity for each PLC Gateway. Each LCN can have up to 20 process networks. Each Data Hiway or Universal Control Network is one process network, so the sum of PLCG Hiway Entities, hiway entities for Data Hiways connected to HGs, and UCNs cannot exceed 20. Please note that the emulated hiway counts as a process network, even though it is not a physical hiway.

Hiway entities are reserved entities; that is, they are entities whose names are reserved for use by the system and begins with a dollar sign. The names are in this form:

`$HIWAYnn`

where nn is the hiway number (01 through 20).

2.2.2 Box/Slot Entities

You must build one box/slot entity for each emulated DHP for which you need to build data points. Each PLCG can have up to eight DHPs.

Like hiway entities, box/slot entities are reserved entities; that is, they are entities whose names begin with a dollar sign, and are reserved for use by the system. The names are in this form:

`$HYnnBxx`

where nn is the hiway number (01 through 20) and xx is the hiway box number (2 and 3 for the PLCG partners (primary and backup) and 8 through 15 for the DHPs).

2.3 ORDER FOR HIWAY AND BOX/SLOT ENTITY BUILDING

The order in which the Hiway and Box/Slot Configuration activities are accomplished is important. To build them, select **HIWAY GATEWAY** on the Engineering Main Menu, which calls up the HG Build Type Configuration Menu. Then select **HIWAY CONFIG** to build the hiway entity and **BOX/SLOT CONFIG** to build the box/slot entities.

Do these activities in this order:

1. **HIWAY CONFIG**
2. **BOX/SLOT CONFIG** for this PLCG pair (HG 2 and 3).
3. **BOX/SLOT CONFIG** for each DHP in use in this PLCG pair (8 through 15).

NOTE

When performing this initial load of a DHP's Box/Slot Configuration, an "A80" Addressing Error will occur. To clear the error, select **Init Addr** from the Hiway Command menu under the Hiway Status Display. Load the DHP Box/Slot Configuration a second time. This time there should be no errors.

The reserved entities that are defined by these tasks must be loaded in the order listed above. This means you must use individual Load commands to load the entities separately in this order, or you must use a Load Multiple command that loads them together in the proper order. For a Load Multiple command, the order in which the entities are loaded is specified by the order that the entity names appear in the selection list file.

You could also load these entities from an exception build source file (.EB file). In the .EB file, entities must be listed in the proper order.

2.4 HIWAY CONFIGURATION

The emulated Data Hiway in a PLCG is not a physical process network, so the PLCG does not have remote or added gateways, as does the Hiway Gateway.

2.4.1 Hiway Number

Each pair of PLCGs (a primary and secondary) on an LCN is assigned a unique process network number (01 through 20). The process network number identifies the hiway database to be used by the PLCG pair.

2.4.2 Hiway Traffic Director Functions

Because there is no physical Data Hiway associated with a PLCG or PLCG pair, the PLCG has complete control over all transactions on the emulated Data Hiway, so a Hiway Traffic Director (HTD) is physically impossible and functionally unnecessary. Therefore, always select **ThisHG** as the value for parameter **HWYHTD** in the hiway entity.

2.4.3 Sequence-of-Events Synchronization

The PLCG does not support sequence-of-events synchronization; therefore, always select Disable as the value for parameter SOESYNCH in the hiway entity.

2.5 BOX/SLOT CONFIGURATION

The type of box on the Data Hiway (HG for the PLCG and DHP for the emulated boxes) is defined in this task, as is each device's address on the hiway. You must configure a box/slot entity for each PLCG in a PLCG pair and you must configure a box/slot entity for each DHP for which you intend to build data points.

NOT RECOMMENDED BY TAC:

Because box/slot configuration takes only a few minutes, you should consider configuring all eight DHPs even if you do not intend to use them. This will make it easier to implement additional DHPs in the future.

After selecting BOX/SLOT CONFIG for the BOXTYPE parameter, select HG to configure a PLCG box/slot entity; and select DHP to configure a DHP box/slot entity.

2.5.1 PLCG Data Hiway Addresses

Assign the first PLCG to hiway address 02 and assign its partner (if present) to hiway address 03. Confirm that the PLCI jumper is in the correct position for redundant or nonredundant PLCG. Refer to the *PLCG Planning, Installation, and Service* manual for details of PLCI jumper options.

2.5.2 Emulated DHP Hiway Address Assignments

Assign the emulated DHPs to hiway addresses 08 through 15.

2.5.3 PLCG and DHP Box Assignment

The PLCG does not support remote or added gateways, as does the HG, so always configure ThisHG as the value for the BOXASSN parameter in both the PLCG box/slot entities and the DHP box/slot entities.

2.5.4 Event Processing

The emulated DHPs support the generation and reporting of events (Process Alarms, Process Changes, etc.). The parameter EVENT PRC should be set to ENABLE to allow routing of these events through the PLCG to the LCN.

2.5.5 Change Detection

Since there are no other Masters on the Emulated Hiway in the PLCG, there is no need to signal or detect changes made by other Masters. The CHNG FLAG parameter should be set to NOT CONFIG.

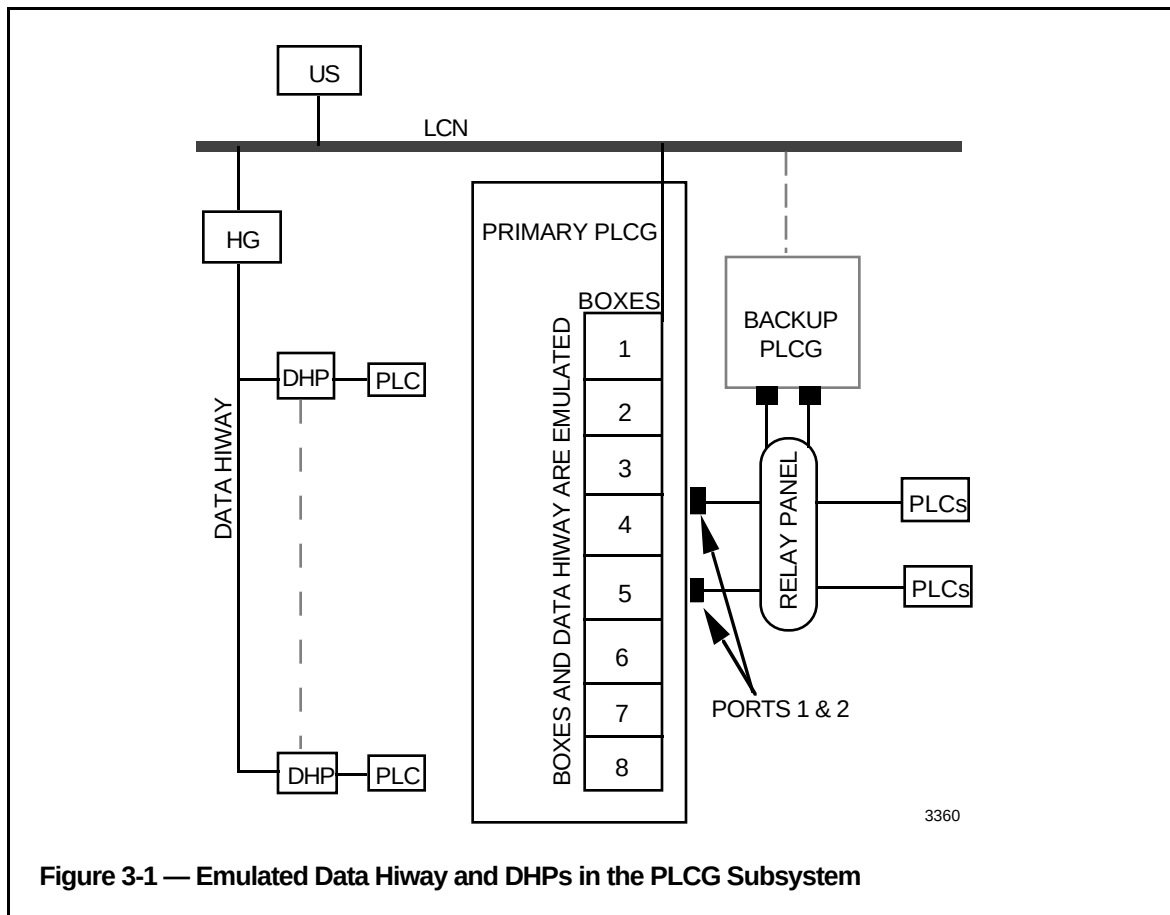
PLCG OPERATIONAL CONSIDERATIONS

Section 3

This section describes operational characteristics of PLC Gateways (PLCGs) that you should consider during PLCG implementation.

3.1 THE PLCG SUBSYSTEM

The PLCG appears to the remainder of the LCN-based system to be a Hiway Gateway (HG) with a Data Hiway and up to eight Data Hiway Ports. It is supported by the HG software with no modifications. Therefore, all standard displays, including Status displays, Group displays, Detail displays, and Engineering Personality displays are the same as if the PLCG was an HG, including all HG, Data Hiway, and DHP terminology.



3.1.1 DHP Point Planning Guidelines

Below is a list of items to consider when implementing DHP points in the PLCG.

1. To avoid potential confusion by operators, do not configure more PC devices in the DHP box/slot entity (\$HYnnBxx) than necessary. Additional devices can be added later, if needed.
2. Plan or map the PLC addresses carefully and use contiguous PLC addresses as much as possible for the most efficient data transfers.
3. To avoid interaction between DHP points:
 - Do not configure more than one DHP point to access the same PLC memory location.
 - Do not use the same slot/subslot for more than one DHP point (tag). Remember that dual input and dual output points require configuration of the first subslot only, with the next higher subslot in the same slot automatically assigned to the second input or output.
4. All points in a DHP slot should have the same PC device index. This will avoid a loss of view to some points accessing data in a good PLC because of a failure in another PLC device. A PC device failure will FAIL all points in any DHP slot for which at least one subslot is configured to access the failed PC device.
5. Note that the DHP box will enter a temporary RESET condition whenever
 - the DHP box/slot entity (\$HYnnBxx) is loaded;
 - a DHP point is loaded to that box (except for load from checkpoint);
 - certain DHP parameters are changed from the point detail display. These parameters include PCADDRxy, SPECIFxy, and PCBITxy, which cause the PLCG to rebuild the scan table for data acquisition.

These operations will cause system alarms and may concern operators. A method to minimize these alarms during configuration is to load DHP points with the DHP box in the RESET state.

NOTE: Loading multiple DHP points with the DHP box RESET is not recommended for a new DHP database, since the point configuration and connection to the PLC device are **not** checked by the DHP box while in the RESET state. Because any bad point could fail the entire slot, debugging point configuration errors after Load Multiple is more difficult.

3.2 LOADING, SAVING, AND RESTORING OF PLCG DATA

3.2.1 PLCG Initial Startup or Restart

PLCGs are loaded with software and data through the Gateways and Interfaces display, which is called up through the System Status display. On the Gateways and Interfaces display, select the node number (or pair of node numbers) for the PLCG that is to start up as the primary PLCG. Then select the appropriate node loading targets and follow the prompts to initiate loading of the PLCG (select **OPERATOR PROGRAM** to load with the operating personality). After the status for the primary PLCG goes to **OK**, you can initiate loading of the backup PLCG. When it is loaded, its status becomes **BACKUP**.

The PLCG is loaded with its software personality image and a data image. The data image consists of previously checkpointed data or null checkpoint (a set of Honeywell-provided checkpoint files with no point data). The data that is loaded from the Gateways and Interfaces display does not include data for the emulated DHPs, which is loaded or restored through the Hiway Status display (for more information, see subsection 3.2.2).

CAUTION

Do not try to load both PLCGs in a PLCG pair concurrently—one of the PLCGs will fail. Complete loading one of the PLCGs as the primary, then the other PLCG can be loaded as the secondary.

For more information about starting or restarting PLCGs, refer to the information about loading of gateways and HG modules in Section 15 of the *Process Operations Manual* in the *Process Operations* binder. For instructions for using null-checkpoints in the initial startup of a PLCG, refer to Task 18 in the *System Startup Guide* (cartridge drives) in the *Implementation/Startup & Reconfiguration - 1* binder.

3.2.2 Initial Startup of a PLCG

A summarized example of 10 steps required to start up a PLCG is presented in this table. Publication references given in this section are here for convenience to the startup effort and some are duplicates of those in the list in subsection 1.2.

STEPS 1 THROUGH 4	NOTES	REFERENCE PUBLICATIONS
1. Physically install the PLCG or redundant PLCG pair; including power, grounding, and LCN cabling.	The connection between the PLCG(s) and the PLC(s) will be discussed in a later step.	* Refer to the publications listed by these numbers below for useful information you may need to accomplish each task: 1, 2, 3, 4, 5
2. Configure the PLCG or PLCG pair as a node or node pair on the LCN.	- The PLCG configures as a Hiway Gateway (HG) and looks exactly like an HG to the operator. - If the PLCG is redundant, you will need to enter the Redundant Node Number ID (the two digit LCN node number). - You will be asked to enter a two digit Hiway number (01 through 20).	* 6, 7, 8, 11, 12
3. Perform a volume configuration to provide checkpoint space on a History Module within your system.	- Although having a checkpoint on your LCN network is not absolutely mandatory (you could store it on removable media), it is highly recommended. - Do not simply add a directory to an existing volume on an HM.	* 6, 7, 8, 11, 12
4. Complete the PLCI board pinning in the PLCG(s) and the PLCG relay panel pinning on the back of the PLCG(s).	- PLCG parity can be either odd or even, no parity is not an option. - If you chose Redundant Allen-Bradley for the relay panel pinning, the port 1 and port 2 pinning of TS2 on the PLCI board must be for identical baud rate and parity. - TS3 jumper settings must match the installed configuration and be the same in both nodes of a redundant pair.	* 1

* See page 3-7 for publication references.

STEPS 5 THROUGH 8	NOTES	REFERENCE PUBLICATIONS
5. Complete the cabling between the PLCG relay panel or I/O card and the PLC(s).	This is a good time to complete any dip switch settings and/or configuration in the PLC(s) you are interfacing to.	* 1
6. Power on the PLCG(s) and load it/them with the HG null checkpoint.	- The null checkpoint files are on volume &HGC, cartridges &CR6 or &C6 - The directory names are &I01 through &I20 corresponding to the appropriate hiway number chosen. - The file name that will be loaded is HG0xxMAS.CP where xx corresponds to the appropriate hiway number chosen. - At the Gateway Status display, the PLCG(s) should show as OK for the primary HG and Backup for the redundant HG (if present).	* 1, 6, 7, 12
7. Build and load the PLCG Hiway Data Point \$HIWAYnn.	The PLCG configures as a Hiway Gateway (HG) and looks exactly like an HG to the operator.	* 6, 7, 8, 11, 12, 13
8. Build and load the PLCG Box Data Point(s) \$HYnnB02 (and \$HYnnB03 if redundant).	IMPORTANT - Do not build and load any of the DHP Box/Slot Data Points until this step is complete. - The PLCG configures as a Hiway Gateway (HG) and looks exactly like an HG to the operator. - For the primary PLCG, the box number has only one valid choice and it is 02. - For the backup PLCG, the box number has only one valid choice and it is 03.	* 6, 7, 10, 11, 12, 13

* See page 3-7 for publication references.

STEPS 9 AND 10	NOTES	REFERENCE PUBLICATIONS
9. Build and load the DHP Box/Slot Data Point(s) \$HYnnB08 to \$HYnnB15	• IMPORTANT - Do not do this by exception building or the use of the load multiple command with IDs. • WARNING - After the initial load of a DHP, there will be an associated A80 addressing error seen from the Hiway Status Display. To clear the error, perform an INIT ADDR command from the HIWAY COMMANDS function. <u>Then reload each and every DHP Box/Slot Data Point again.</u> DO NOT PROCEED WITHOUT COMPLETING THIS PROPERLY. • Each emulated DHP should come up in the RESET mode. Performance of an ENABLE PROCESSING command should make each DHP go to an OK state.	* 6, 7, 10, 11, 12, 13
10. Build and load data Point(s).	• The DHP box must be in BASIC control state to load points. Points will not load if DHP is in FULL control state. • One point is enough to cause communications to start between the PLCG(s) and a PLC. Thus it is recommended that only one point be built and loaded before going further. • Perform a demand checkpoint now. This way you will have something to go back to in case of a power failure or equipment malfunction. It is best to do a save using the ALL BOXES target to insure saving both the HG and DHP box portions of the database. NOTE: When deleting a DHP point with the DELETE ENTITY function of the Data Entity Builder (DEB), be sure to first set to zero (0) the PCADDRxy and SPECIFxy (Modbus), or PCBITxy (Allen-Bradley digital) parameters for the DHP point. The DELETE ENTITY function does NOT delete point data from the emulated DHP box memory, and the point will continue to be read from the PLC as long as the PCADDRxy and SPECIFxy or PCBITxy parameters are non-zero. This is commonly referred to as a “ghost point.”	* 10, 11, 12, 13, 14

PUBLICATION TITLE	BINDER TITLE	
1. PLCG Planning, Installation, and Service	Implementation/PLC Gateway	
2. PLCG Specification and Technical Data	Implementation/PLC Gateway	
3. LCN Site Planning	System Site Planning	
4. LCN System Installation	LCN Installation	
5. LCN System Checkout	LCN Installation	
6. System Startup Guide - Cartridge Drives	Implementation/Startup & Reconfiguration - 1	
7. System Startup Guide - CD-ROM	Implementation/Startup and Reconfiguration - 1	
8. System Startup Guide - Zip Disk	Implementation/Startup and Reconfiguration - 1	
9. PLC Gateway Implementation Guidelines	Implementation/PLC Gateway	
10. Data Entity Builder Manual	Implementation/Engineering Operations - 1	
11. Engineer's Reference Manual	Implementation/Startup & Reconfiguration - 2	
12. PLC Gateway Forms	Implementation/PLC Gateway	
13. PLC Gateway Control Functions	Implementation/PLC Gateway	
14. PLCG Parameter Reference Dictionary	Implementation/PLC Gateway	
NOTE: Release 530, and later, software updates are distributed only on Zip Disks and CD-ROM.		

3.2.3 Saving of PLCG and DHP Data

After its initial startup, you can begin to build the database for your PLCG(s), including the data points and hiway and box/slot entities described in Section 2. These entities (points) are built with the Data Entity Builder, which you activate by selecting HIWAY GATEWAY on the Engineering Main Menu.

As you load points into the PLCG and during operation, you should periodically checkpoint (save) your PLCG and DHP data on an HM or removable media (cartridges or floppies). Use the SAVE DATA target on the Hiway Status display to request a demand checkpoint. You can enable automatic, periodic checkpointing to an HM through the AUTO SAVE target on the Gateways and Interfaces display (you should disable periodic checkpointing while you are building data points). For more information on checkpointing, refer to Section 21 of the *Engineer's Reference Manual* in the *Implementation/Startup & Reconfiguration - 2* binder.

During initial configuration, we recommend that, in addition to checkpoints being saved automatically on an HM, you periodically request demand checkpoints through the **SAVE DATA** target on the Hiway Status display to be stored on removable media. This will serve as backup data if the HM should become unavailable. We do not recommend copying checkpoint files from an HM to removable media, because if more than one cartridge is needed, the data for one unit may be split between cartridges or floppies, and such data cannot be reloaded.

After installation is complete, we recommend disabling automatic checkpointing on a PLCG/DHP, and doing manual checkpoint (save) after configuration changes. Because automatic checkpoint does not save data for a DHP in any state other than OK, it is possible to have incomplete checkpoint files due to communication link errors between PLCG and a PLC, or due to accidental errors during configuration changes.

NOTE

It is possible to save data for individual DHPs (boxes), but before you can do so, you must have saved a complete checkpoint for the PLCG and all its emulated DHPs. Because the DHPs are not really individual physical boxes, but are integral parts of the PLCG, it is probably best to select **ALL BOXES**, rather than individual boxes. This ensures that all data changes in the PLCG and all DHPs are saved.

3.2.4 Restoring PLCG and DHP Data

PLCG and DHP data can be restored from checkpoint files through the **LOAD DATA** target on the Hiway Status display. A successful load replaces any data in the PLCG and DHP(s) with data from the checkpoint files. These loads can be directed to any box (DHP), or you can select **ALL BOXES** to direct the load to all DHPs.

NOTE

After reloading the PLCG personality, all configured DHPs will have A80 Address Errors. These errors must be cleared using the **INIT ADDR** command in the Hiway Command Menu from the Hiway Status Display. The configured DHPs will now be in "Reset." The DHP data can now be restored.

To accept a load, the PLCG and DHP status on the Hiway Status display must be **OK**, and the box control state must be **BASIC**. When a load is completed successfully, the word **RESTORED** appears adjacent to the box's status indicator.

3.2.5 Recovering from Checkpointing and Loading Errors

The following error messages may appear on the Gateways and Interfaces display or on the Hiway Status display when you attempt to save PLCG checkpoint and box checkpoint data, when you attempt to load a PLCG or boxes, or when you attempt to save box checkpoint data:

- **FILE ERROR**—See subsection 3.2.5.2.
- **BAD REQUEST**—A request was made for a box configured as **AddedHG** or **RemoteHG**, and not **ThisHG**. The PLCG and all emulated DHPs must be configured as **ThisHG**.
- **CONFLICT**—The PLCG is busy with another save or load operation.
- **NO VOLUME**—Either no cartridge is mounted in the drive specified, or the volume doesn't exist on the HM, or no checkpoint files were found on the specified medium (cartridge or HM).
- **BAD STATE**—The box was not in a proper state to accept the command. See subsection 3.2.2.
- **???????????**—A communication error or some other error occurred that prevents the Universal Station from displaying the correct status.

Some operations might terminate because of an error, such as a cartridge not mounted, or a bad medium, and a message such as **FAIL ; HGnn** appears. This doesn't mean the PLCG failed, it's the operation that failed.

3.2.5.1 Recommended Actions for Error Recovery

1. Use the **SCREEN PRINT** function to print a copy of the Gateway Status display or the Hiway Status display. If boxes are involved, you may need to print more than one display to document both the HG status and box status.
2. Check the Real-Time Journal print out(s) on the printer. This information will be useful if you need to consult with Honeywell about the problem. Error messages that result from these operations are in this form:

```
D$ CP CHPNT xx yy zz
```

- 5001 through 5063, where the last two digits are the number of the box that is involved, or 60nn, where nn is the HG file number, or 0 = the system "crashed."
- Provides a cause-of-error indication.
- Indicates the internal HG module name.

2958

3.2.5.2 Recovery from a FILE ERROR

A FILE ERROR message is caused by one of the following:

- An attempt to load a box that was not previously checkpointed.
- Needed checkpoint files were not found on the medium (cartridge or HM) that was specified.
- A hiway error occurred as access to a box was attempted.
- On a SAVE DATA operation, the medium was full (too few files formatted) or the directory was too small to accept the data.
- An error occurred as access to the medium was attempted.

You should use the Catalog (LS or CAT) command in the Engineering Personality's Utilities to check the number of available files and sectors for the PLCG checkpoint data. Use the command's detail (-d) option. See *Command Processor Operation* in the *Implementation/Engineering Operations - 1* binder.

You can determine if checkpoint data for a specific box is on the medium by using the Utilities List File Attributes (LS) command to find information for the HG0hhbbr.CP file for the box. It may be useful to check the time stamps that are in the catalog data to see if an incompatible time stamp may be causing a problem. For example, box data should not be older than PLCG data.

3.2.6 Recovery from DHP Point Configuration Errors

Occasionally, DHP slot (card) failures or PC device failures occur due to incorrect point configuration or improper procedure during configuration changes. Below is a list of common problems and recovery procedures.

SYMPTOM: DHP slot (card) failures or PC device errors occur after deleting DHP points or moving points to a different DHP box, slot, or subslot.

CAUSE: This is often caused by improper deletion of the DHP point before moving it. The parameters PCADDRxy, SPECIFxy, and PCBITxy must be set to zero (0) prior to deleting a DHP point to remove it from the PLCG scan. If this is not done, the old point becomes a “ghost” which continues to be scanned by the PLCG. Such ghost points are saved in the checkpoint file, so the only way to remove them is by loading a point to the same DHP box/slot/subslot and making a new checkpoint file.

SYMPTOM: DHP slot (card) failures or PC device errors occur after changing DHP slot type or making a slot NotConfigured.

CAUSE: To properly change DHP slot type, all the points in the slot must be deleted. Be sure to delete the points properly by first setting PCADDRxy, SPECIFxy, or PCBITxy parameters to zero (0). If the slot appears empty of points, but the slot type change is still unsuccessful or causes errors, there are probably ghost points in the “unused” subslots. Be sure the subslots are clear by loading a dummy point to every subslot and setting the PCADDRxy, SPECIFxy, or PCBITxy parameters to zero (0), then deleting the dummy point(s).

Worst Case Scenarios: When slot or PC device failures elude your best efforts to clean up configuration errors, more drastic measures may be required to load a clean database. This involves clearing the database in the DHP box or entire PLCG, then restoring from Exception Build (*.EB) files or Intermediate Data Files (*.IDF). Follow these steps:

1. Save the existing database by executing SAVE, ALL BOXES to NET and to removable media. Do *not* use these files except in an emergency, since the checkpoint is suspected of containing ghost points. Also note that failed DHP boxes will not be saved, so you may need to retain older checkpoint files made prior to the DHP failure.
2. Backbuild the PLCG database to create *.EB files for desired points.
3. Use the DEB (Data Entity Builder) or DocTools to create selection lists (*.EL files) for all DHP points. Put the Hiway (\$HIWAY) and box/slot (\$HYnnBxx) entities in a separate list from the DHP points.
4. From the DEB, Print System Entities to create the *.EB files, using the *.EL files just created.
5. Document the existing database using DocTools.
6. Build a DocTools query with the following parameters:
 BOXNUM, SLOTNUM, INPTSSLT, OUTSSLT, PNTBOXIN, PCADDRI1, SPECIFI1, PCBITI1, PCADDRI2, SPECIFI2, PCBITI2, PNTBOXOT, PCADDRO1, SPECIFO1, PCBITO1, PCADDRO2, SPECIFO2, PCBITO2
 These are the parameters that provide the unique correlation between DHP box/slot/subslot location and PLC memory.
7. Save this query for later use.
8. Run the query, then sort the results using BOXNUM SLOTNUM and INPTSSLT in the Field Name List.
9. Output the results to a file and save for later use.
10. Prepare for Loss of View and Loss of Control, then clear the existing database. Depending on the severity of the problem, it may suffice to clear only the DHP database with the CONFIG CLEAR command for the affected DHP. Worst case, shutdown both PLCG nodes (if redundant) and cycle power on the nodes.
11. Reload the PLCG by following subsection 3.2.2 “Initial Startup of a PLCG” in this Implementation Guidelines manual. Be sure to load in the order shown: PLCG node on LCN, then \$HIWAY entity, then PLCG entities (\$HYnnB02/3) on the hiway, then DHP boxes (\$HYnnB08/15) on the hiway, then DHP points. Use Exception Build in the DEB to load the box/slot and DHP point entities from the *.EB files created earlier. DO NOT reload the point data from checkpoint files, since these are suspected of containing the ghost points which caused the original problems.
12. Compare the new PLCG database to the old one to check for errors. Use the prebuilt DocTools Query saved earlier to document the PLCG database and save results to a file, then compare the old and new DocTool query results.
13. Save the new PLCG database by doing SAVE, ALL BOXES to NET. Do it twice to update both file sets.

This completes the process and the ghost points have been eliminated.

3.3 USE OF HOLDING REGISTERS IN MODICON PLCs

The holding registers in Modicon and Modicon-compatible PLCs can be used as sources of PLCG digital input data, thereby increasing throughput by transferring blocks of digital data rather than single bits. To do so, a Digital Input or Digital Composite point's subslot address in parameter INPTSSLT specifies the bit to input from the holding register, as follows:

Subslot number in INPTSSLT	16	15	14	-----	02	01
Holding register bit number	16	15	14	-----	02	01

3.4 REPORTING BY EXCEPTION IN ALLEN-BRADLEY PLCs

To reduce the communication load presented to Allen-Bradley PLCs by the PLCG, the PLC's exception-reporting feature can be used. If this option is chosen after the initial PLC data has been scanned by the PLCG, it expects PLCs to notify it of significant changes in the PLC data by sending unsolicited inputs to the PLCG.

To activate reporting by exception, add two to the port address in the PCnPORT parameter (n = PLC number, 1 through 8). A port address of 1 or 2 specifies that the PLC is connected to ports 1 or 2, respectively, and that continuous scanning of the PLCs is configured. A port address of 3 or 4 specifies that the PLC is connected to ports 1 or 2, respectively, and that reporting by exception from the PLCs is configured. RBE message can also be accepted for points which are scanned by PLCG.

For RBE, the PLC must be programmed to detect database changes and send updates to the PLCG. The PLC database is generally divided into blocks, and when a change is detected in a block, the entire block is sent to the PLCG. If the message transfer is successful, no further effort is required until the next data change is detected. Retry handling is determined by the PLC programmer.

To avoid undetected loss of communication from the PLCs, keep-alive PLC addresses and bits must be configured in the box/slot entity for the DHP serving each Allen-Bradley PLC (parameters PCnALIVE AND PCnALVBT, where n = PLC numbers 1 through 8). If communication of a keep-alive bit fails, the PLCG places all points in the target PLC on continuous scan until the data is reestablished, and all data points that reference that PLC fail. Once communication is reestablished and new values are obtained, reporting by exception resumes.

If this option is chosen, the ladder-logic program in the PLC must support reporting by exception and must initiate all such reporting. Because periodic scanning of the PLC data is not occurring, changes by Universal Station operators and user-written programs must be reported back to the PLCG by the PLC. If it is not, the operators or user-written programs will not be able to see the status changes.

3.4.1 Communication to Allen-Bradley PLC-5

To communicate with a PLCG, the PLC-5 must use the PLC-2 emulation mode with byte addressing. Allen-Bradley full duplex (DF1) protocol is used on an EIA RS-232-C link.

To communicate with a PLCG, it is recommended that you use an integer file (type “N”). Typical implementation involves creation of an intermediate integer file in PLC-5, used only for PLCG communication. All files in the PLC-5’s memory are addressed with a decimal number (9-999). Ladder logic must be implemented to transfer data in the intermediate integer file to and from the appropriate PLC-5 files actually used for PLC-5 processing. Performing block moves is one method of transferring data.

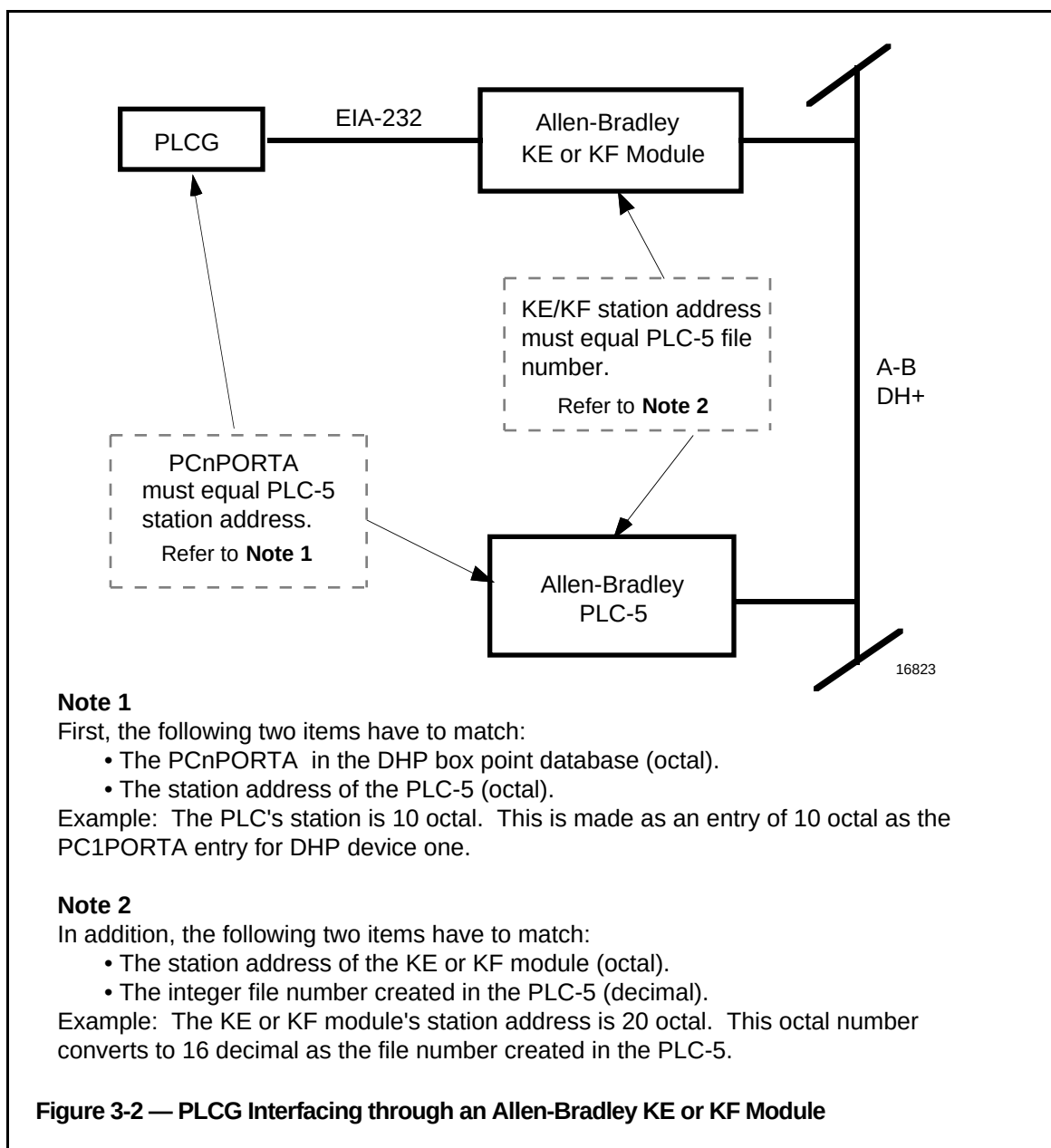
The PLCG and Allen-Bradley full duplex (DF1) protocol has no mechanism for specifying a PLC-5 file number. Therefore, the PLC-5 integer file number must be the decimal equivalent to the octal DH+ address of the A-B device physically linked to the PLCG (see the box slot configuration example below). All stations on the communications link are numbered in octal. Each station should have a unique station address.

Use PLCG parameter PCn TYPE (model type) = APLC, for correct binary value format, when interfacing PLC-5 in PLC-2 emulation mode. Selection of other model types results in communications of values in BCD format.

The PCnPORTA parameter value must be the same as the DH+ address of the target PLC-5 processor (see the box slot configuration example below).

3.4.2 Box Slot Configuration Example

In the figure below, we will use an example of a PLCG interfacing through an Allen-Bradley KE or KF module by way of the Allen-Bradley Highway Plus (DH+) to a single Allen-Bradley PLC-5.



3.4.3 Process Point Example

When building any point in the PLCG, use PCADDRIn or PCADDROn equal to the address (element) of the data in the PLC-5 integer file that was created for PLCG access. For example, if the data for a PLCG analog input is in PLC-5 location N16:22, use PCADDRI1 = 26 octal for the PLCG analog input point. Note that for PCADDRIn and PCADDROn a value of 0 (zero) is illegal for the PLC and will cause the point to not be scanned by the PLCG. Values on the PLCG point are in octal while those in the PLC are in decimal.

In addition, when building a digital point in the PLCG, the PLCG parameter PCBITIn or PCBITOn specifies which bit position a PLCG digital point occupies within the PLC-5 integer file register. Once again values on the PLCG point are in octal while those in the PLC are in decimal.

For more details on PLCG parameters, refer to the *PLCG Parameter Reference Dictionary* (PL09-540 or equivalent).

For pinning options on the PLCI board, refer to the *PLCG Planning, Installation, and Service* manual (PL02-500 or equivalent), as well as recommendations on Allen-Bradley interface module settings.

PLCG PERFORMANCE CONSIDERATIONS

SECTION 4

4.1 FACTORS LIMITING PERFORMANCE

Although it is a very fast processor of data, the PLCG has finite computational capacity as its upper limit. If the processing of input/output data is restricted by the PLCG's inability to perform required computations, it is known as "compute bound." Limiting factors at the other extreme are baud rate and PLC I/O delays. If computations must wait for data to be received or transmitted, the condition is called "I/O bound." As baud rate is increased and/or PLC I/O delays are decreased, performance improvement is linear and usually proportionate. The performance improvement degrades, however, as compute limits are approached.

Statistics are maintained at various levels which can be used to monitor the performance of the PLCG. See *PLCG Planning, Installation, and Service* manual, subsection 4.3.2 for details and subsection 4.3 for how to display them.

4.2 EFFECTS OF DATABASE COMPOSITION

The performance of a data base chiefly composed of analog inputs, analog outputs and/or counters (numeric points) is primarily limited by baud rate and PLC performance. This is because numerics are easily processed in less than the 16 bit times required to transmit them to the PLCG. The performance of a data base which is predominantly digital inputs and digital outputs (Boolean points) places a greater burden on computational power because only one bit time is available for processing.

4.3 ADJUSTING A PLCG FOR PERFORMANCE

Of the factors affecting PLCG performance, the most significant is baud rate. As a rule of thumb, doubling the baud rate will double performance. If you are already running at the maximum baud rate, consider dividing the data base between the PLCG's two ports, if possible. This has the same effect as doubling the baud rate. Pinning the PLCI board to double buffer Data Requests (pipelined operation) will also improve performance. See *PLC Gateway Planning, Installation, and Service* manual, subsection 3.2.3.4, for pinning information.

At the PLC end, a dedicated communications module should be used if supported by the PLC. These modules usually contain a dedicated processor that utilizes a high-speed access mechanism, such as DMA, to access the PLC data. The PLCG requests data in blocks of up to 64 words. Unused data elements in the block will be read and discarded. For example: two 30-word blocks separated by 4-unused words will be collected as a single 64-word block. This is faster than two 30-word requests. If a 60-word block is preceded or followed by unused words, only the 60-word block will be requested. By using consecutive storage elements within the PLC, the amount of excess data requested by the PLCG can be reduced. Contiguous addressing of the PLC data elements provides optimum performance because Modbus and Allen-Bradley protocol specify start address and number of elements to include in any transfer.

After point load or reconfiguration of some parameters, the PLCG will sort all PLC database addresses before starting to request data, so the order of the points in the emulated DHP is not critical. As long as the PLC data elements are contiguous, the desired result will be achieved.

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